
UNIVERSITY OF SOUTHERN CALIFORNIA

EE 483: Digital Signal Processing

Lecture 12: Signal Reconstruction, Zero-Order Hold, and Oversampling

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1 Aliasing and Signal Reconstruction

Aliasing: high frequencies can look like low frequencies.

1.1 Signal Reconstruction (for Bandlimited Signals)

Let $X(j\Omega) = 0$ for $\forall |\Omega| \geq \frac{\pi}{T}$. Then

$$x(t) = \frac{1}{2\pi} \int_{-\pi/T}^{\pi/T} X(j\Omega) e^{j\Omega t} d\Omega.$$

Let $\omega = \Omega T \Leftrightarrow \Omega = \frac{\omega}{T} \Rightarrow d\Omega = \frac{1}{T} d\omega$:

$$x(t) = \frac{1}{2\pi T} \int_{-\pi}^{\pi} X(j\frac{\omega}{T}) e^{j\frac{\omega}{T} t} d\omega.$$

From Lecture 11,

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j\Omega) \Big|_{\Omega=\frac{\omega}{T}+\frac{2\pi k}{T}} = \frac{1}{T} X(j\frac{\omega}{T}) \quad \text{for } |\omega| \leq \pi,$$

so

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\frac{\omega t}{T}} d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[\sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \right] e^{j\frac{\omega t}{T}} d\omega \\ &= \sum_{n=-\infty}^{\infty} x[n] \left[\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\frac{\omega t}{T} - j\omega n} d\omega \right]. \end{aligned}$$

Reconstruction formula.

$$x(t) = \sum_{n=-\infty}^{\infty} x[n] \frac{\sin(\pi(\frac{t}{T} - n))}{\pi(\frac{t}{T} - n)}.$$

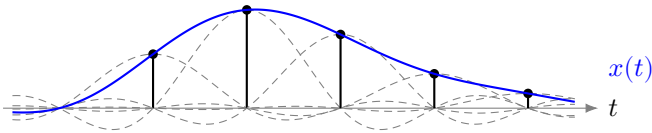
Remark

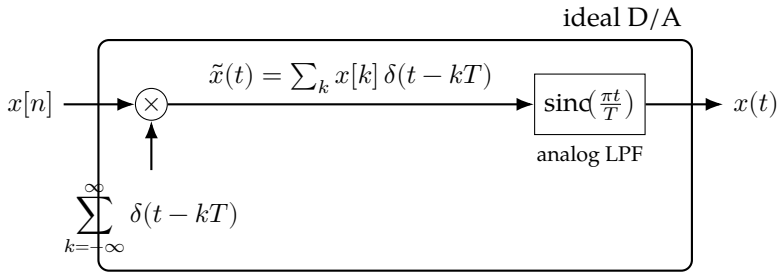
(addendum to orig lecture note).

With $a = \frac{t}{T} - n$, the bracket is $\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j\omega a} d\omega = \frac{e^{j\pi a} - e^{-j\pi a}}{2\pi j a} = \frac{\sin(\pi a)}{\pi a}$. In this course $\text{sinc}(x) = \frac{\sin x}{x}$, so the kernel is $\text{sinc}(\pi(\frac{t}{T} - n))$: it equals 1 at $t = nT$ and 0 at every other sample time, so $x(nT) = x[n]$.

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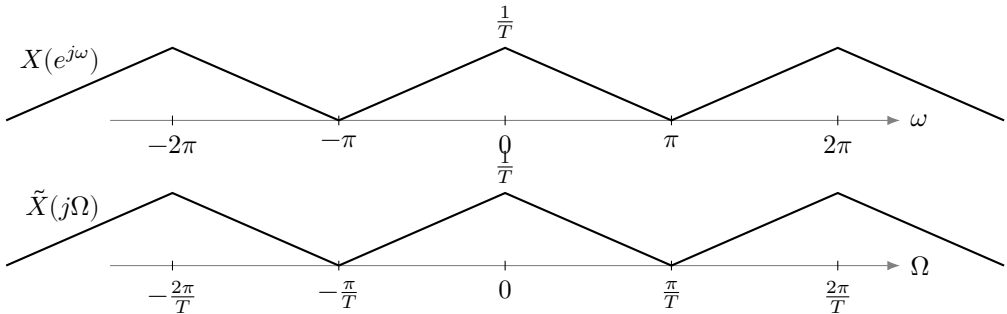
1.2 Sinc Interpolation: Discrete Time to Continuous Time





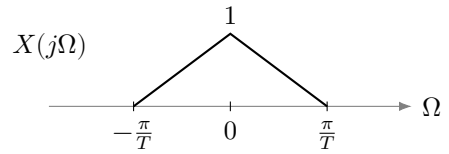
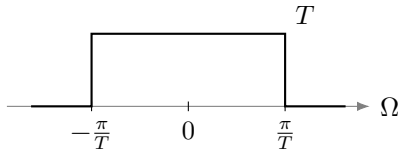
What is the CTFT of $\tilde{x}(t)$?

$$\begin{aligned}\tilde{X}(j\Omega) &= \int_{-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} x[k] \delta(t - kT) \right] e^{-j\Omega t} dt \\ &= \sum_{k=-\infty}^{\infty} x[k] e^{-j\Omega kT} = X(e^{j\omega}) \Big|_{\omega=\Omega T}.\end{aligned}$$



3 2 Zero-Order Hold (ZOH)

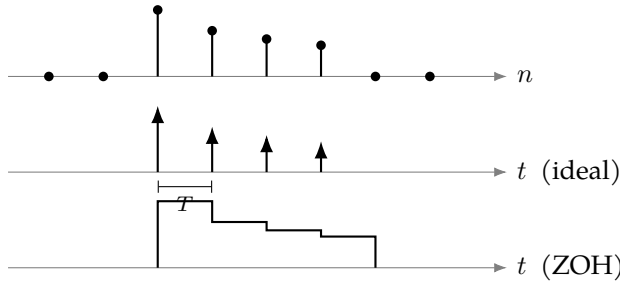
Note: $\text{sinc}(\frac{\pi t}{T})$ has CTFT T on $|\Omega| < \frac{\pi}{T}$ (and 0 outside), so the analog LPF passes the central copy of $\tilde{X}(j\Omega)$ scaled by T , which is $X(j\Omega)$:



Zero-order hold (ZOH).

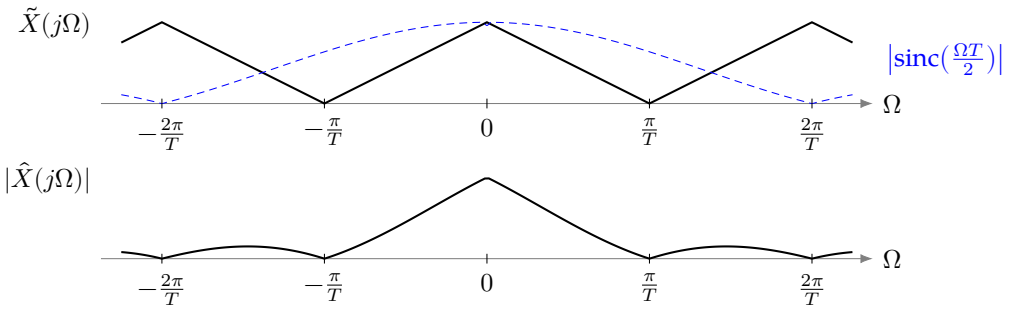
Instead of outputting a stream of Dirac delta functions, output each sample value and hold it constant until the next sample comes:

$$\hat{x}(t) = \sum_{n=-\infty}^{\infty} x[n] r(t - nT), \quad r(t) = \begin{cases} \frac{1}{T}, & 0 \leq t \leq T, \\ 0, & \text{else.} \end{cases}$$



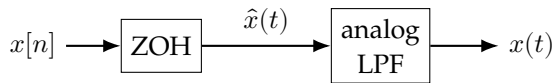
2.1 Spectrum of the ZOH Output

$$\begin{aligned} \hat{X}(j\Omega) &= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} x[n] r(t - nT) e^{-j\Omega t} dt \\ &= \sum_{n=-\infty}^{\infty} x[n] \underbrace{\int_{-\infty}^{\infty} r(t - nT) e^{-j\Omega(t-nT)} dt}_{\frac{1}{T} \cdot T e^{-j\frac{\Omega T}{2}} \text{sinc}\left(\frac{\Omega T}{2}\right)} e^{-j\Omega nT} \\ &= \left[\sum_{n=-\infty}^{\infty} x[n] e^{-j\Omega nT} \right] e^{-j\frac{\Omega T}{2}} \text{sinc}\left(\frac{\Omega T}{2}\right) \\ &= X(e^{j\omega}) \Big|_{\omega=\Omega T} \cdot \text{sinc}\left(\frac{\Omega T}{2}\right) e^{-j\frac{\Omega T}{2}}. \end{aligned}$$



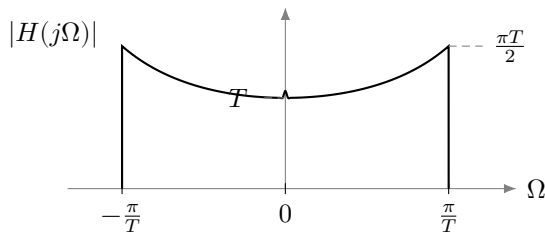
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2.2 Compensating Analog LPF



$$H(j\Omega) = \begin{cases} \frac{T e^{j\Omega T/2}}{\text{sinc}(\Omega T/2)}, & |\Omega| \leq \frac{\pi}{T}, \\ 0, & \text{else.} \end{cases}$$

$\Rightarrow$ **eliminated the need for Dirac delta functions!**



- We still need non-causal, infinite analog filters.
- **Practically:** we'll use a causal approximation (delay), and we'll also approximate the analog frequency response (inaccuracy).

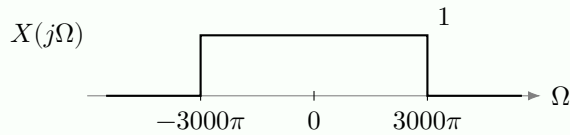
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3 Oversampling

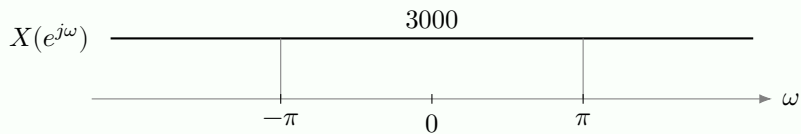
Note: we can sample faster than the Nyquist rate (sampling faster $\Rightarrow$ smaller T).

Example.

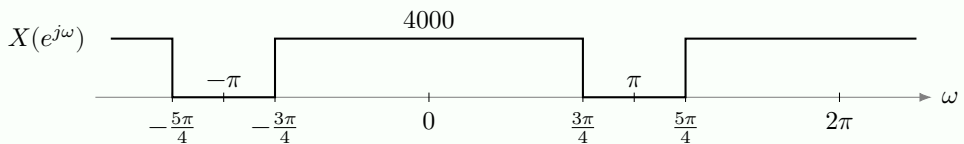
$X(j\Omega) = 1$ for $|\Omega| \leq 3000\pi$ and 0 else, so the Nyquist rate is 3000 Hz.



$T = \frac{1}{3000}$ s (sampling at the Nyquist rate): $X(e^{j\omega}) = 3000$ for all ω .



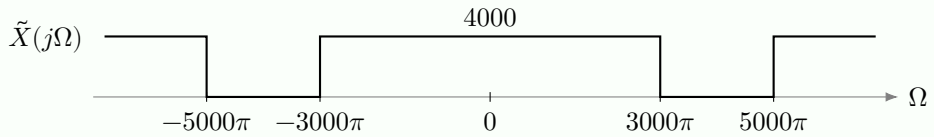
Let $T = \frac{1}{4000}$ s $\Rightarrow$ sampling rate 4000 Hz:



Oversampling might be good: it relaxes our constraints on the analog LPF.

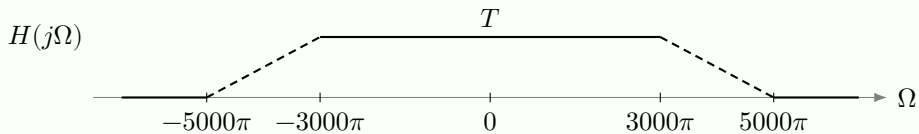
Example (continued).

The output of the ideal D/A before filtering:



The analog LPF must satisfy

$$H(j\Omega) = \begin{cases} T, & |\Omega| \leq 3000\pi, \\ 0, & |\Omega| \geq 5000\pi, \\ \text{don't care,} & \text{else.} \end{cases}$$



Much easier than designing a sharp cutoff frequency!

Readings

- Monson H. Hayes, *Schaum's Outline of Digital Signal Processing*, 2nd ed., Ch. 3.5 (upsampling, downsampling, sample rate conversion)
- Sanjit K. Mitra, *Digital Signal Processing: A Computer-Based Approach*, 4th ed., Secs. 3.8, 3.10